

Semester – VI

Course Code	Course Name	Credits
AIML2603	Artificial Intelligence Lab	1

Contact Hours			Credits Assigned			
Theory	Practical	Tutorial	Theory	Practical	Tutorial	Total
-	-	-	-	02	-	01

	Theory					Term Work / Practical/Oral			Total
Internal Assessment				End Sem Exam	Duration Of End Sem Exam	Term Work	Pract.	Oral	
Test	Continuou s Evaluation	Attendance	Total Internal						
10	15	05	30	70	-	-	2 Hours	100	

Prerequisites

- Understanding of Programming Concepts, Familiarity with Python and Prolog.

Course Outcome

1. Apply various techniques for solving problems using prolog programming Language.
2. Implement the elementary searching algorithms.
3. Practice different production rules in water jug problem
4. Design different problems such as monkey banana problem
5. Simulate the various graph traversing algorithms.
6. Design and develop an Expert System that operates in a realistic problem domain and communicate effectively in a team or individual.

Objectives

1. To conceptualize the basic ideas and techniques of AI
2. To distinguish various search techniques and to make students understand knowledge representation and planning.
3. To become familiar with the basics of NLP and highly professional languages.
4. To familiarize yourself with Hybrid systems and to build expert systems.

Detailed Syllabus

1	Write a program to calculate the factorial of a number using prolog
2	WAP to find the length of a list using prolog
3	WAP to reverse the list using prolog
4	Write a program to solve the N queens' problem
5	Solve any problem using depth first search and Breadth First Search.
6	Solve any problem using Best first 'search
7	Solve 8-puzzle problem using best First search
8	Solve water jug problem by giving all the production rules.
9	Solve Monkey banana problem.
10	Implement A* search
11	Design an expert system with any realistic problem domain in pyhton

Recommended Books:

1. Stuart J. Russell and Peter Norvig, "Artificial Intelligence a Modern Approach Second Edition" Pearson Education.
2. Samir Roy and Chakraborty, Introduction to soft computing, Pearson Edition.
3. S. Rajasekaran and G. A. VijayalakshmiPai "Neural Networks, Fuzzy Logic and Genetic Algorithms", PHI Learning.
4. N. P. Padhy, Artificial Intelligence and Intelligent Systems, Oxford University Press
5. "Introduction to AI and Expert Systems", D.W. Patterson, PHI,
6. "Introduction to Expert Systems", A Peter Jackson, AWP, M.A.
7. "Artificial Intelligence - an Engineering Approach", R.J. Schalkoff, McGraw Hill Int. Ed., Singapore,

Semester - VI

Course Code	Course Name	Credits
CSE2615	Cryptography and System Security Lab	1

Contact Hours			Credits Assigned			
Theory	Practical	Tutorial	Theory	Practical	Tutorial	Total
.	02	-	-	01	-	01

Lab							Term Work /Practical/Oral			Total
Internal Assessment					End Sem Exam	Duration Of End Sem Exam	Term Work	Prac	Oral	
Performance	Lab Record	Viva	Attendance	Total Internal						
10	10	05	05	30	70	3 Hours	-	35	35	100

Course Outcome

- A Cryptography and System Security Lab serves as a crucial platform for advancing the field of cryptography and evaluating security systems. By conducting research and development activities, it enables experts to explore new cryptographic techniques, algorithms, and protocols to enhance data protection. The lab provides a controlled environment for testing and evaluating security systems, identifying vulnerabilities, and developing effective countermeasures. Through its efforts, the lab contributes to strengthening the security of digital systems, protecting sensitive information, and ensuring the integrity and confidentiality of data.

Course Objectives

- To be able to research and develop advanced cryptographic algorithms, protocols, and techniques: The lab aims to explore new methods and approaches to encryption, key management, and cryptographic protocols. It focuses on developing more secure and efficient cryptographic algorithms to protect data confidentiality, integrity, and authenticity.
- To be able to analyze and evaluate existing cryptographic systems: The lab strives to assess the strength and weaknesses of current cryptographic systems by conducting thorough analysis and evaluations. It aims to identify vulnerabilities, potential attacks, and limitations of existing cryptographic mechanisms and propose improvements or alternatives.
- To be able to design and implement secure cryptographic solutions: The lab aims to design and implement secure cryptographic solutions tailored to specific applications and use cases. It focuses on developing encryption algorithms, secure key exchange protocols, digital signatures, and other cryptographic primitives that can be integrated into real-world systems and applications.

Detailed Syllabus

Sr. No.	Experiment Title
1.	Design and Implementation of a product cipher using Substitution and Transposition ciphers